



Koneru Lakshmaiah Education Foundation

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## STUDENT CASE STUDY EXECUTION REPORT

|                |                           |
|----------------|---------------------------|
| Department     | FED                       |
| Sub'ect        | DSA-2                     |
| Academic Year  | 1-III 2025-26<br>Semester |
| Facult Name    | Mohd.Razia                |
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| Roll Number    | 2520090149                |
| Branch         | CSIT                      |
| Course Outcome | CO-5                      |

Case Study Topic: Streaming Request Scheduling using Priority Queue

### Work Done Summary:

In this case study, we implemented a Priority Queue to manage streaming requests in the Media Vault platform. User requests were assigned priorities based on subscription level, streaming quality, and request time. The Priority Queue ensured that high-priority users received faster service and reduced streaming delays.

Various streaming requests were inserted, processed, and removed according to their priorities. The system efficiently handled large numbers of requests while maintaining smooth streaming performance.

### Implementation Details:

- 1) Created a Priority Queue for streaming request management.
- 2) Assigned priorities based on subscription plans.
- 3) Inserted user streaming requests into the queue.
- 4) Processed requests according to priority order.
- 5) Implemented request removal after streaming completion.
- 6) Compared Priority Queue performance with normal queue processing.
- 7) Verified efficient scheduling and reduced waiting time.

## Result: Screenshots / Output

```
1  import java.util.PriorityQueue;
2  class StreamingRequest implements Comparable<StreamingRequest> {
3      int priority;
4      String userName;
5      StreamingRequest(int priority, String userName) {
6          this.priority = priority;
7          this.userName = userName;
8      }
9      @Override
10     public int compareTo(StreamingRequest other) {
11         return other.priority - this.priority;
12     }
13 }
14 public class MediaVaultPriorityQueue {
15     public static void main(String[] args) {
16         PriorityQueue<StreamingRequest> pq = new PriorityQueue<>();
17         pq.add(new StreamingRequest(5, "Premium User"));
18         pq.add(new StreamingRequest(4, "Gold User"));
19         pq.add(new StreamingRequest(3, "Basic User"));
20         pq.add(new StreamingRequest(2, "Free User"));
21         System.out.println("Streaming Requests Processing Order:");
22         while (!pq.isEmpty()) {
23             StreamingRequest request = pq.poll();
24             System.out.println(
25                 request.userName +
26                 " (Priority: " +
27                 request.priority + ")");
28         }
29     }
30 }
31 }
```

Output:-

```
1  Streaming Requests Processing Order:
2  Premium User (Priority: 5)
3  Gold User (Priority: 4)
4  Basic User (Priority: 3)
5  Free User (Priority: 2)
```

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The Priority Queue successfully managed streaming requests and ensured faster service for premium users. The system reduced response time and improved overall streaming efficiency.

GitHub Link: <https://github.com/snehiths399-byte/MediaVault--Digital-content-streaming-and-Analytics-system/blob/main/Priority%20Queue>

Student Signature

Faculty Signature